

DATA MINING & MACHINE LEARNING

LOCOMOTION MODE RECOGNITION FROM EXOSKELETON-EMBEDDABLE SENSORS

Data Mining & Machine Learning Project
M.Sc. In Bionics— Academic Year 2025–2026

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PART 1

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PART 1 — TECHNICAL

Problem, Pipeline & Results

Problem definition · feature engineering · model selection
hyperparameter optimization · evaluation · test-set results

Problem Definition & Dataset

The Problem

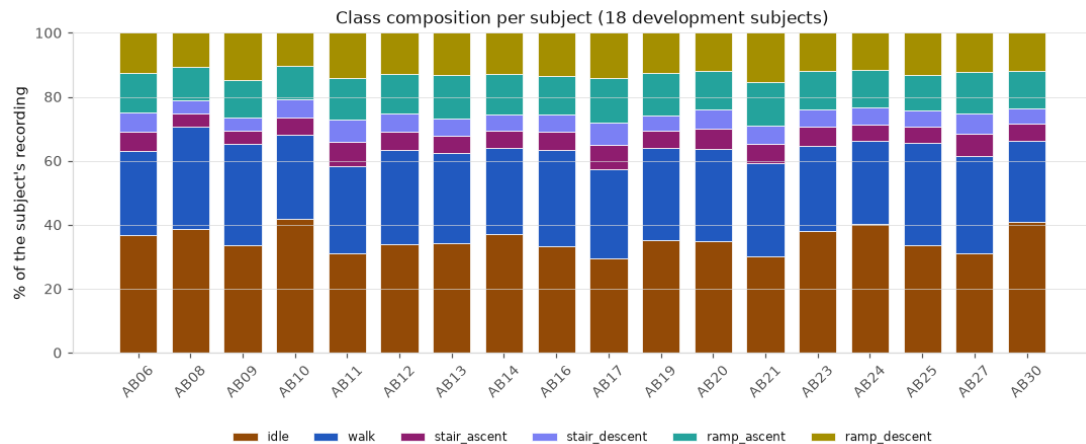
- Recognizing the **locomotion mode** of a person from wearable IMU and goniometer sensors, as needed to control a lower-limb exoskeleton.
- Multiclass classification – 6 classes : *idle* , *walk* , *stair_ascent* , *stair_descent* , *ramp_ascent* , *ramp_descent* .
- Assistive robotic devices (exoskeletons) must understand the user's intended movement to adapt torque/control— locomotion-mode recognition from embeddable sensors (IMU + goniometers, no cameras or lab motion-capture) is a prerequisite for adaptive, real-world assistive control.
- **Strong class imbalance** combined with high inter-subject variability in movement patterns, requiring models that generalize to unseen subjects (subject-independent / LOSO validation) rather than just to unseen trials.

Dataset at Glance

Task	<i>Multiclass classification</i>
Dataset	<i>Camargo et al. (2021), EPIC Lab — 22 subjects</i>
Target	<i>Locomotion mode</i>
Features	<i>555: 15 statistics × 37 channels</i>
Missing values	<i>0.0044% of samples overall</i>
Class balance	<i>35.32% / 28.3% / 12.85% / 12.45% / 5.67% / 5.41%</i>
Split	<i>18 subjects (18-fold LOSO) / 4 held-out (test)</i>
Metric(s)	<i>F1-Macro</i>

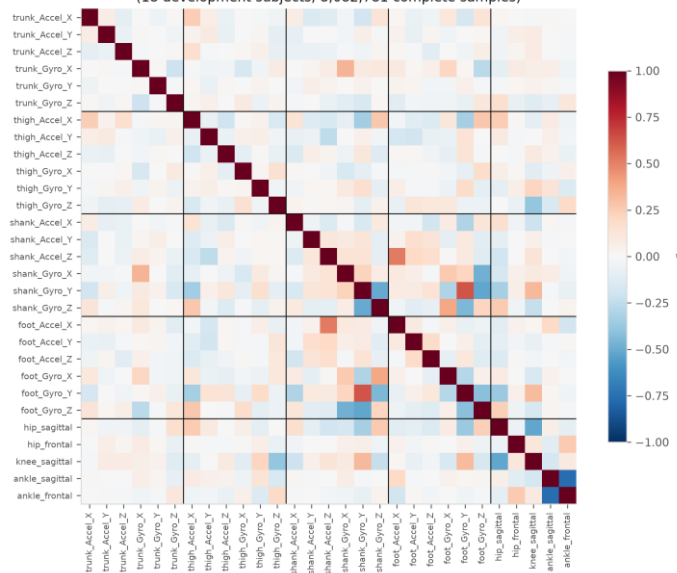
PROBLEM

Exploratory Data Analysis



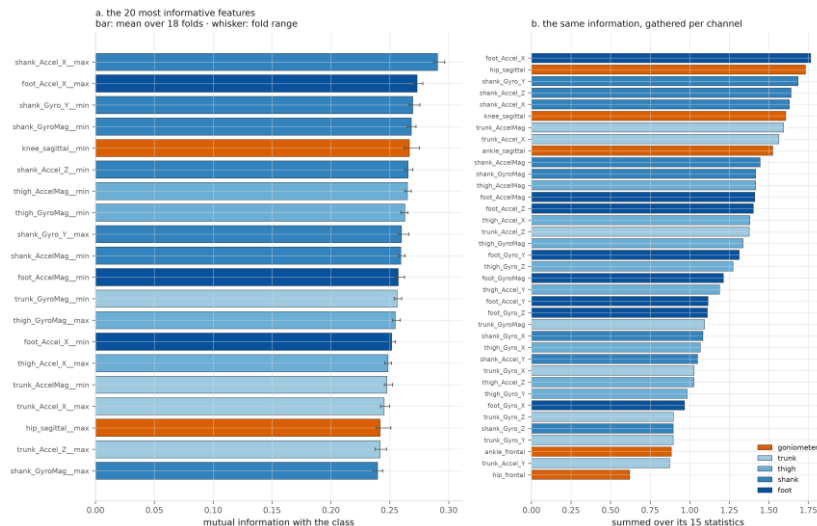
- **Low correlation** between channel from different sensors.
- **Low number of missing values** across the dataset.
- **High difference** in class distribution between subjects.

Figure 2 - Pearson correlation between the 29 channels (18 development subjects, 8,082,781 complete samples)



Class	Missing values
stair_descent	0.00703%
idle	0.00240%
walk	0.00069%
ramp_ascent	0.00001%
stair_ascent, ramp_descent	0%

Feature Engineering & Selection



Why These Features?

- **Domain justification:** 8 magnitude channels ($|\text{accel}|$, $|\text{gyro}|$ per segment) added because they're rotation-invariant — reduces inter-subject variability from sensor mounting orientation.

Engineering

- **New features:** per-window (250 ms with 125ms overlap) statistical + spectral summaries per channel — 15 statistics $\times$ 37 channels = 555 features
- **Normalization:** *StandardScaler*, fit only on the training fold, applied to val/test — no leakage.
- **Imputation:** *none* — windows containing any invalid sample are dropped entirely.

Selection Method

- **Method:** filter — mutual information (MI) between each feature and the label .
- **Criterion:** MI score, ranked once per training fold .
- **Validated:** 18-fold subject-independent (LOSO) CV sweep.
- **Final k-best:** $k = 350$, choose in the hyperparameter tuning.

ML Pipeline: Avoiding Data Leakage & Bias



Data Leakage Prevention:

- **Initial division** of the subject 18 – 4 Held out
- All **choices** were based on the 18 development subjects.
- All preprocessing fitted only on training folds

Class Imbalance Handling:

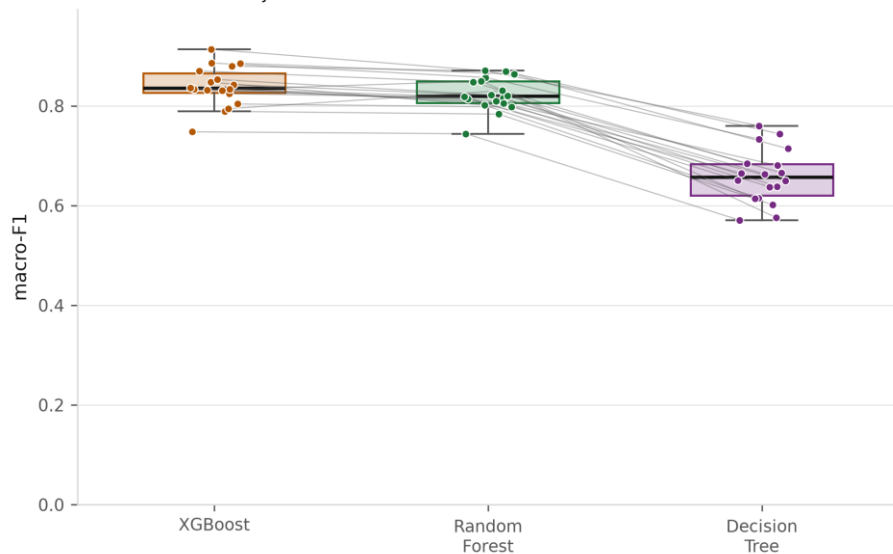
- **Undersampling** of idle windows to 40% of their count, mainly for compute cost.
- Remaining imbalance handled via `class_weight="balanced"` (Decision Tree), `class_weight="balanced_subsample"` (Random Forest), and `compute_sample_weight("balanced", y_train)` passed as `sample_weight` (XGBoost).
- All done only for the training.
- Chosen metric: macro-F1

Metric Justification:

- Macro-F1 used for class imbalance
- FP and FN error cost variable for each class.
- Precision and recall alone each emphasize a different error type – false positives vs. false negatives – whose relative cost varies by class, hence macro-F1.

Model Selection & Hyperparameter Optimization

Leave-one-subject-out scores, 18 subjects held out in turn
each line is one subject: the forest is ahead on 18 of 18 of them



Key Decisions

Candidate models	Decision tree – Random Forest - XGBoost
CV strategy	LOSO 18 - Fold
Selection criterion	Macro-F1
Chosen model	XGBoost slightly better overall results
Search strategy	Grid search
Search space	k feature, learning rate, $n_{estimators}$
Best config	$k=350$, $max_leaves=63$, $learning_rate=0.3$, $n_{estimators}=600$
Statistical test	Wilcoxon test + Holm correction

RESULTS

Test-Set Results

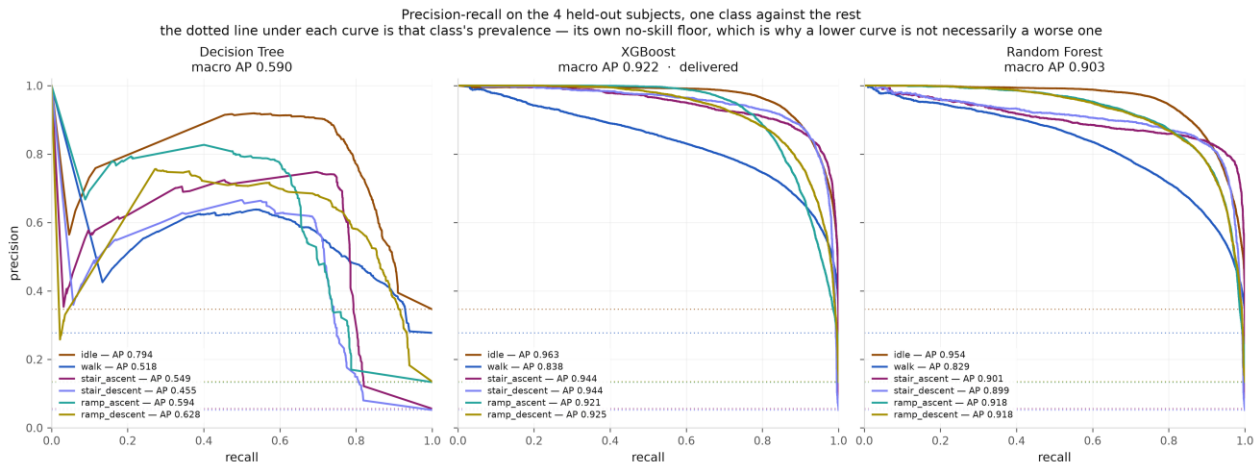
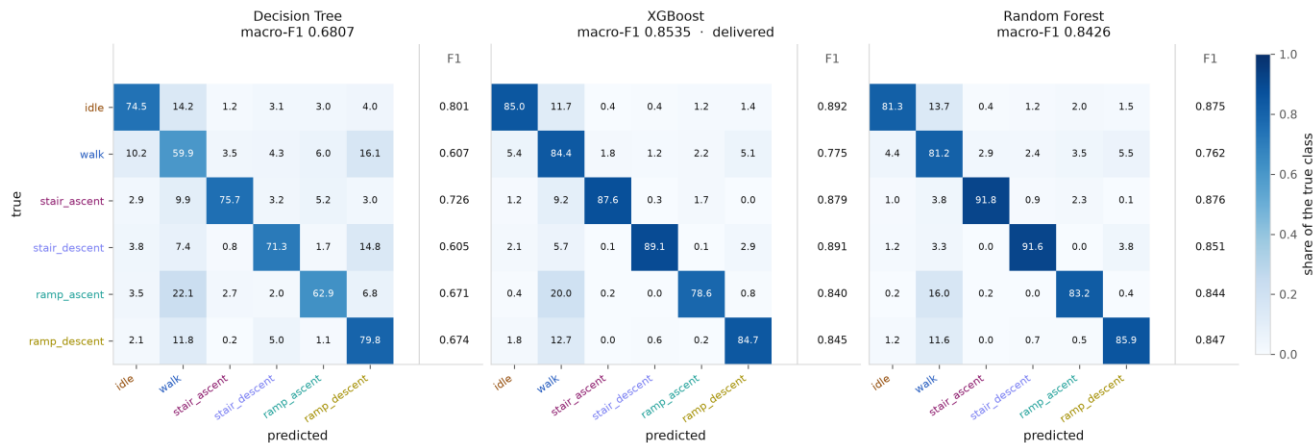
Key Metrics LOSO vs Held out

$0.839 \pm 0.04 \rightarrow 0.854$
XGBoost

$0.823 \pm 0.03 \rightarrow 0.843$
Random Forest

$0.659 \pm 0.05 \rightarrow 0.681$
Decision Tree

Confusion matrices on the 4 held-out subjects, rows normalised
each row is one true class and sums to 100: the diagonal is that class's recall in per cent, and the column on the right is its F1

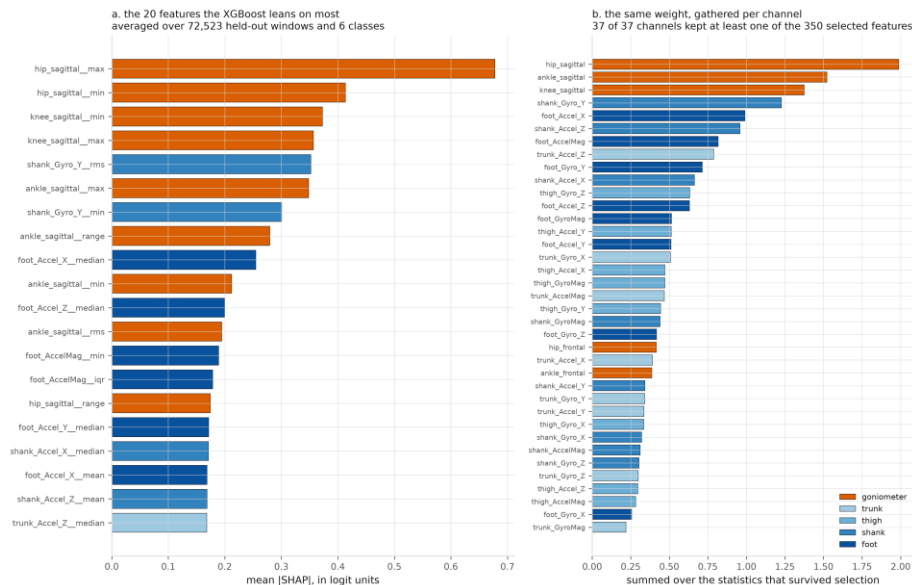


PART 2 — CRITICAL ANALYSIS

Explainability & Critical Discussion

How does the model work? · What did it get wrong and why?
Limitations · future directions · domain impact

Global Explainability: What Drives the Model Overall?

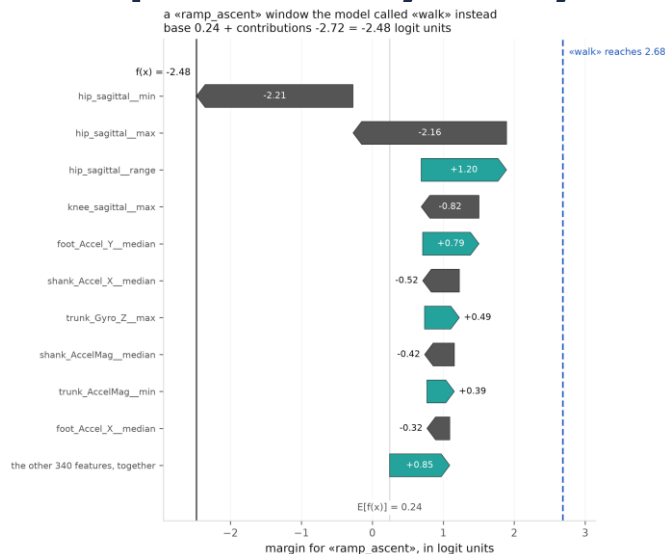


Interpretation:

- **TreeSHAP** – optimized for tree ensembles, it allow to study the importance of each feature in the classification.
- The model leans on the **sagittal goniometers**. This follow the biomechanics, as all the tasks are done in the sagittal plane.
- The ranking is completely different from the MI ones.
- Invariant channels are not so important compared to the absolute.

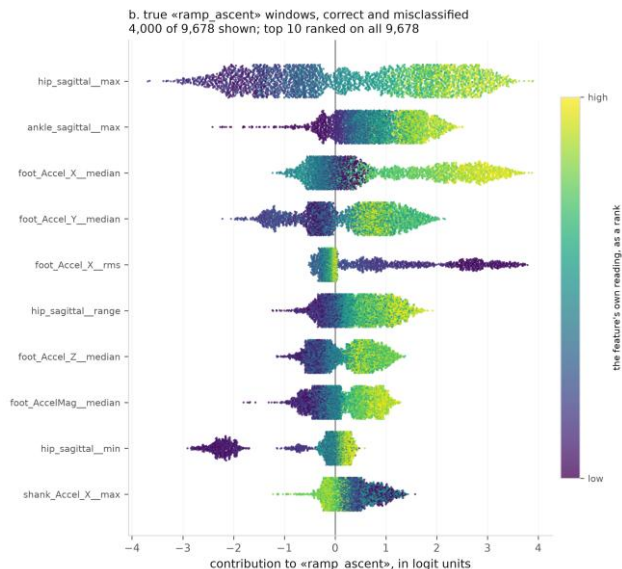
Family	Feature	Expected	Final
Absolute	164	46.9 %	70.6 %
Invariant	186	53.1 %	29.4 %

Local Explainability: Why This Specific Prediction?



Misclassified window:

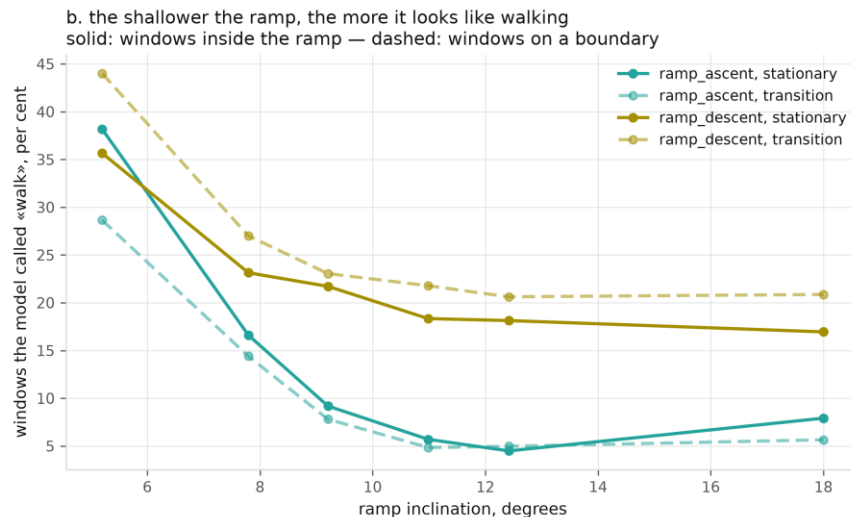
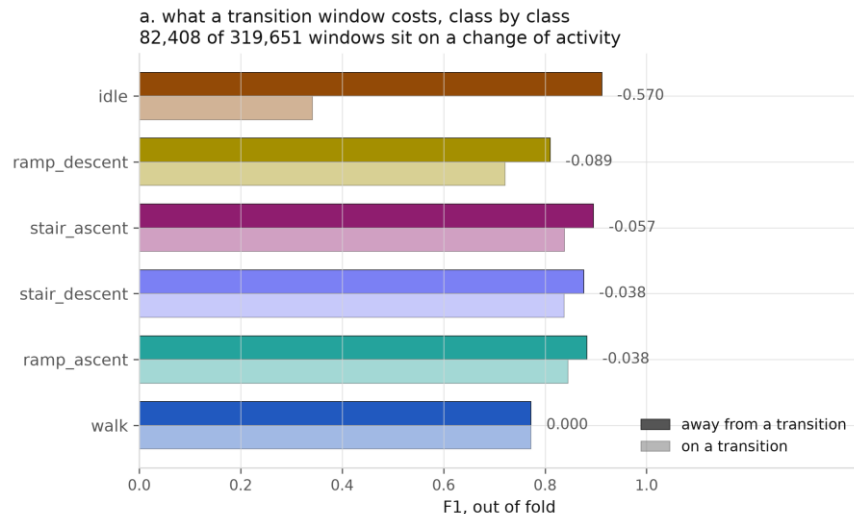
- Base 0.24, margin -2.48: driven by hip_sagittal min/max (-2.21/-2.16).
- Low hip_sagittal pushed the score away from ramp_ascent.



Beeswarm plot:

- Beeswarm: high hip/ankle_sagittal_max raises the score, low lowers it — the same pattern behind the error above.
- Matches the global SHAP ranking — probably not an outlier case.
- Biomechanically plausible: less range of motion makes a ramp look flatter, more walk-like.

Error Analysis: What Did the Model Get Wrong and Why?



Reverse-Engineering Errors:

- idle drops hardest on transitions: F1 0.913→0.345 (−0.570) — residual movement reads as motion.
- ramp_descent also drops (−0.089); walk unaffected (−0.000) — catch-all role.
- Ramp→walk share falls as incline rises: ~38%/22% at 5.2°, single digits above 11°.
- Systematic, not random: transition-heavy idle, low-incline ramp windows.
- idle→walk is safety-critical: risk of unwanted assistance while stationary.

Domain Impact, Limitations & Future Directions

Domain Impact

- Real-time recognition of locomotion mode (walk/idle/ramp_ascent/etc.) to drive an assistive/rehabilitation device (e.g. an exoskeleton) .
- Automatic switching between assistance modes (e.g. "provide support " vs. "no assistance because").
- The **idle**→**walk** confusion is the most safety-critical: it could trigger unwanted assistance.
- With a held-out macro-F1 of 0.8535, the system can support semi-autonomous control, but the idle/walk confusion argues for keeping a safety margin rather than full unsupervised trust.

Limitations

- Only 4 held-out subjects — the positive generalization gap is encouraging but cannot rule out residual overfitting.
- Right-side sensors only, sagittal plane only ;the scenario is constrained by design to a purely rehabilitation context with only six locomotion modes.
- Hard/discrete labels for transitions, while real boundaries are ambiguous; the 250ms/125ms window was chosen for low computational cost, not optimized for discriminability.
- Single dataset, no real-world deployment test, population restricted to the rehabilitation context.

Future Directions

- Test on a less constrained population (not limited to rehabilitation)
- Frontal/transverse plane, left-side sensors, exploring other window lengths to see whether they limit real-world performance.
- Target the labeling itself for transitions (e.g. finer-grained transition classes, or a soft/uncertain output instead of a hard label) to reduce idle-related errors.

Thank You

Conclusions:

- We classified six locomotion modes (including walk, idle, ramp_ascent) from wearable sensor data in a rehabilitation scenario, using sagittal-plane goniometer features in short time windows.
- XGBoost achieved the highest macro-F1 in both LOSO cross-validation and on held-out test subjects (0.8535 vs. 0.8389 validation), with no evidence of leakage — confirmed as the delivered model.
- SHAP shows the model relies on biomechanical signal (sagittal-plane goniometer features dominate importance), and explains the main confusions — a biomechanical angle effect for the ramp classes, and a transition-labeling effect for idle, with the safety-critical idle→walk confusion being the most concerning.
- With only 4 held-out subjects, residual overfitting can't be fully ruled out; the natural next step is a deployment study on a broader population, alongside improving the labeling of transitions itself (e.g. finer-grained classes or soft/uncertain outputs) to reduce idle-related errors.